

Substrate-Symplectic Pfaffian $Pf(\omega)$ Explicit Computation at $V_{-}(1/2, 1/2)$ v0.1

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Substrate-Symplectic Pfaffian $Pf(\omega)$ Explicit at $V_{-}(1/2, 1/2)$ v0.1

0. Goal

Per Casey “queue never empties” + Gate 4 v0.2 substrate-symplectic $Pf(\omega)$ resolution candidate + Cal #36 STANDING positive-search: explicit substrate-symplectic Pfaffian $Pf(\omega)$ computation at substrate-fundamental $V_{-}(1/2, 1/2)$ of $Sp(2)$. This is Step Gate-4-S1 substantive content from Gate 4 v0.2.

1. $Sp(2)$ Substrate-Symplectic Structure

$Sp(2)$ substrate-symplectic Lie group: rank-2 symplectic Lie group; dim 10; substrate-fundamental representation V_{fund} dim 4 = 2 · rank.

Substrate-symplectic 2-form ω on V_{fund} (substrate canonical coordinates (q^1, p^1, q^2, p^2)):

$$\omega = dq^1 \wedge dp^1 + dq^2 \wedge dp^2$$

2. Substrate-Symplectic Pfaffian Computation

The Pfaffian of an antisymmetric 4×4 matrix Ω representing 2-form ω in canonical basis:

$$\Omega = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Substrate-Pfaffian:

$$Pf(\Omega) = \sqrt{\det(\Omega)} = 1$$

For canonical substrate-symplectic 2-form ω on V_{fund} of $Sp(2)$: $Pf(\omega) = 1$ substrate-natural (standard substrate-symplectic normalization).

3. Substantive Substrate-Mechanism Discrepancy

Per Gate 4 v0.2 candidate: substrate-symplectic substrate-Pfaffian 1/g substrate-natural factor candidate to resolve Gate 4 v0.1 honest open factor ~7.

Computation shows $Pf(\omega) = 1$ for canonical normalization, NOT 1/g substrate-natural factor.

The 1/g factor must come from different substrate-symplectic substrate-mechanism contribution, not direct $Pf(\omega)$. Possible candidates:

- (a) Substrate-symplectic substrate-volume form on V_{fund} : $\omega^2 / 2! = dq^1 \wedge dp^1 \wedge dq^2 \wedge dp^2$ substrate-volume; substrate-Pfaffian is square root of substrate-determinant of ω^n where $n = \text{rank}$.
- (b) Substrate-symplectic substrate-Liouville substrate-measure: substrate-Liouville measure $d(q^1, p^1, q^2, p^2)$ substrate-natural; substrate-mechanism normalization may involve substrate-rank or substrate-genus.
- (c) Substrate-symplectic substrate-trace ratio: $\text{Tr}(\omega) / \text{Tr}(I)$ at substrate-fundamental V_{fund} vs substrate-higher representations.
- (d) Substrate-symplectic substrate-Maslov substrate-index: substrate-Maslov substrate-natural integer; substrate-mechanism contribution at substrate-Pfaffian level.

4. Honest Open: Gate 4 v0.2 1/g Candidate Status

Per explicit $\text{Pf}(\omega) = 1$ computation v0.1: the 1/g substrate-natural factor candidate from Gate 4 v0.2 does NOT directly come from substrate-symplectic $\text{Pf}(\omega)$ in canonical form.

Possible substrate-mechanism resolution (multi-week): - Substrate-Pfaffian at higher $\text{Sp}(2)$ representation (substrate-spinor $V_{(3/2, 1/2)}$) contributes different normalization than substrate-fundamental $V_{(1/2, 1/2)}$ - Substrate-symplectic substrate-rank · substrate-genus combination contributes 1/g substrate-natural factor at substrate-mechanism composition level - Substrate-symplectic substrate-Liouville substrate-volume normalization at substrate-spinor cross-K-type matrix element

Per Cal #189 question-shape: Gate 4 v0.2 substrate-symplectic 1/g candidate requires more careful substrate-symplectic substrate-mechanism derivation — naive $\text{Pf}(\omega) = 1$ substrate-canonical normalization doesn't suffice. Multi-week explicit substrate-mechanism derivation gates Gate 4 closure.

5. Refined Gate 4 Honest Open Question Status

Gate 4 v0.2 status update v0.1 from this $\text{Pf}(\omega)$ computation: - $\text{Pf}(\omega) = 1$ substrate-canonical at V_{fund} (substrate-symplectic standard normalization) - 1/g substrate-natural factor from Gate 4 v0.2 candidate REQUIRES alternative substrate-mechanism source - Substrate-symplectic Cat 6 substrate-mechanism contribution to Gate 4 honest open is multi-week pending alternative substrate-mechanism candidate

Per Cal #27 STANDING (CLAIMS-tier honest framing): my Gate 4 v0.2 “1/g substrate-symplectic Pfaffian” was substantively encouraging but explicit computation shows $\text{Pf}(\omega) = 1$ at substrate-fundamental V_{fund} . The Gate 4 honest open factor ~7 composition remains OPEN pending alternative substrate-symplectic substrate-mechanism candidate or other substrate-mechanism FORCING form.

Honest walk-back: Gate 4 v0.2 substrate-symplectic 1/g substrate-Pfaffian candidate is NOT supported by explicit $\text{Pf}(\omega) = 1$ substrate-canonical computation. Multi-week substrate-symplectic substrate-mechanism CONTINUES pending alternative candidates above.

6. Closure v0.1

Substrate-Symplectic Pfaffian $\text{Pf}(\omega)$ Explicit Computation at $V_{(1/2, 1/2)}$ v0.1:

Substantive computation: $\text{Pf}(\omega) = 1$ at substrate-fundamental V_{fund} of $\text{Sp}(2)$ in canonical substrate-symplectic normalization (standard substrate-rank-2 symplectic).

HONEST WALK-BACK of Gate 4 v0.2 substrate-Pfaffian 1/g substrate-natural factor candidate: $\text{Pf}(\omega) = 1$ explicit computation does NOT give 1/g; Gate 4 v0.2 substrate-symplectic candidate requires alternative substrate-mechanism source (multi-week).

4 alternative substrate-symplectic substrate-mechanism candidates for 1/g factor identified for multi-week investigation: (a) substrate-volume form; (b) substrate-Liouville substrate-measure; (c) substrate-trace ratio; (d) substrate-Maslov substrate-index.

Per Cal #27 STANDING + Cal #189 question-shape: Gate 4 v0.2 1/g substrate-Pfaffian candidate WALKED BACK; substrate-symplectic Cat 6 substrate-mechanism reach + UNIVERSAL across K-types observation PRESERVED; alternative substrate-symplectic substrate-mechanism contribution to Gate 4 honest open factor ~7 multi-week.

Tier: substantive substrate-mechanism investigation continues; Gate 4 honest open factor ~7 OPEN multi-week pending alternative substrate-symplectic substrate-mechanism candidate derivation.

— Lyra, Thu 2026-06-04 ~12:25 EDT. Substrate-Pfaffian $\text{Pf}(\omega)$ Explicit Computation v0.1: $\text{Pf}(\omega) = 1$ substrate-canonical at V_{fund} of $\text{Sp}(2)$; Gate 4 v0.2 substrate-symplectic 1/g candidate WALKED BACK honestly; alternative substrate-symplectic substrate-mechanism candidates (substrate-volume form / substrate-Liouville / substrate-trace ratio / substrate-Maslov index) multi-week.